Seguridad red

Mikel Egaña Aranguren

mikel-egana-aranguren.github.io

mikel.egana@ehu.eus

Universidad del País Vasco
Euskal Herriko Unibertsitatea
BILBOKO INGENIARITZA ESKOLA
ESCUELA DE INGENIERÍA DE BILBAO

Seguridad red

https://doi.org/10.5281/zenodo.4302267

https://github.com/mikel-egana-aranguren/EHU-SGSSI-01



Modelo OSI (Open Systems Interconnection)

Layer			Protocol data unit (PDU)	Function[27]
Host layers	7	Application	Data	High-level protocols such as for resource sharing or remote file access, e.g. HTTP.
	6	Presentation		Translation of data between a networking service and an application; including character encoding, data compression and encryption/decryption
	5	Session		Managing communication sessions, i.e., continuous exchange of information in the form of multiple back-and-forth transmissions between two nodes
	4	Transport	Segment	Reliable transmission of data segments between points on a network, including segmentation, acknowledgement and multiplexing
Media layers	3	Network	Packet, Datagram[28]	Structuring and managing a multi-node network, including addressing, routing and traffic control
	2	Data link	Frame	Transmission of data frames between two nodes connected by a physical layer
	1	Physical	Bit, Symbol	Transmission and reception of raw bit streams over a physical medium

https://en.wikipedia.org/wiki/OSI_model

TCP/IP

<table>
<tr><th>OSI Model</th><th colspan="6">TCP/IP Protocol Suite</th></tr>
<tr><td>Application
Presentation
Session</td><td>File Transfer

FTP
TFTP</td><td>Web Browser

HTTP</td><td>Email

SMTP
IMAP
POP3</td><td>Remote Login

Telnet</td><td>Name Resolution

DNS</td><td>IP Address

DHCP</td></tr>
<tr><td>Transport</td><td colspan="3">Transmission Control Protocol
TCP</td><td colspan="3">User Datagram Protocol
UDP</td></tr>
<tr><td>Network</td><td colspan="3">Internet Protocol
IP</td><td colspan="3">ARP
ICMP</td></tr>
<tr><td>Data Link</td><td colspan="2">Ethernet</td><td>Token Ring</td><td>FDDI</td><td colspan="2">WAN Protocols</td></tr>
<tr><td>Physical</td><td colspan="6">Copper Twisted Pair
Fiber Optic
Wireless</td></tr>
</table>

https://www.cdebyte.com/news/889

CIDR (Classless Inter-Domain Routing)

Asignar direcciones IP y enrutamiento en redes

Ejemplo: 192.168.1.0/24

red: 192.168.1.0 - 192.168.1.255 / host o subred 192.168.1.1 - 192.168.1.254

Mascara de red (Routers de una red a otra): 255.255.255.0

Captura de trafico de red

Laboratorio Wireshark

Segmentación de red

Segmentacion física

Segmentacion lógica mediante VLAN (Capa 2 OSI): agrupar dispositivos por funcionalidad

Segmentacion lógica mediante subredes (Capa 3 OSI): dividir una red en subredes más pequeñas (Rango de direcciones IP)

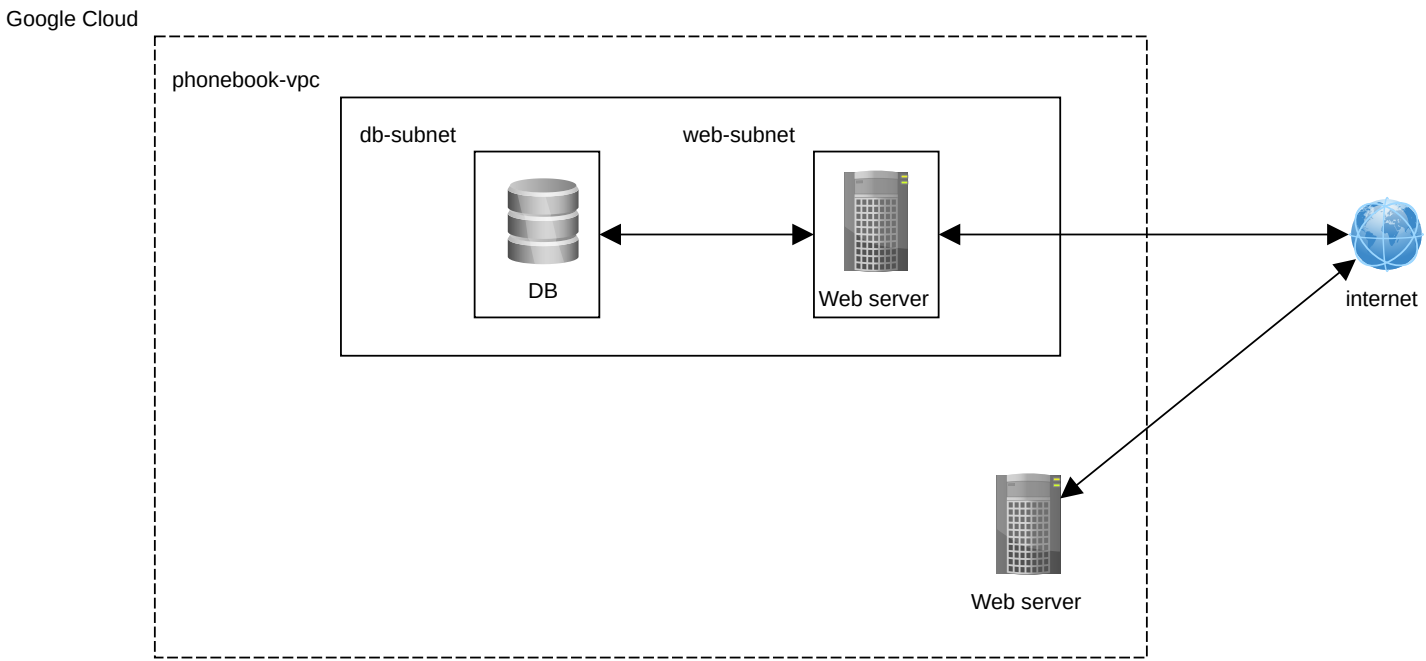
VPC (Virtual Private Cloud)

- Aislamiento: la red está lógicamente separada de otros usuarios de la nube
- Control de IPs: elegir los rangos de direcciones IP, subredes y tablas de rutas
- Gateways: gateways de internet y NAT controlan el tráfico entrante y saliente

VPC Google Cloud

Los objetivos de las reglas de firewall se definen mediante tags (ej. http-server, https-server) en vez de bloques CIDR

VPC Google Cloud



VPC Google Cloud

Laboratorio VPC Google Cloud